

Hammad Ahmad

AI/ML Researcher | Trustworthy NLP · Retrieval-Augmented Generation · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

RESEARCH PROFILE

AI/ML researcher with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication (Springer). My work centres on reliable, transparent language systems: retrieval-augmented generation grounded in knowledge graphs to reduce hallucination, with rigorous benchmarking, alongside predictive modelling on high-dimensional real-world health data. I am seeking a PhD to develop this into rigorous research on trustworthy, knowledge-grounded NLP.

RESEARCH INTERESTS

Knowledge-grounded and retrieval-augmented NLP; hallucination reduction through structured grounding; evaluation frameworks for RAG systems; interpretable machine learning on high-dimensional health data.

EDUCATION

University of Bradford

Sep 2024 - Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics (Merit)

Bradford, UK

- **Dissertation:** FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation. Supervisors: Dr Tillal Eldabi and Dr Irfan Mehmood.
- Designed and evaluated a hybrid architecture combining retrieval-augmented generation, a Neo4j knowledge graph and a locally served Mistral 7B model to reduce hallucination in UK financial-regulation question answering, grounded in the FCA Handbook, PRA Rulebook, FRC standards and statutory instruments.
- Evaluated on a 110-item benchmark of questions, document tasks and case scenarios using RAGAS plus custom citation-precision and legal-completeness metrics: 0.82 source accuracy and 0.81 citation quality, improving on a standalone LLM baseline, complemented by qualitative stakeholder evaluation.
- Selected modules: Artificial Intelligence and Data Science (79), Business Data Analytics (79), Responsible AI: Ethics, Law and Governance (75).

COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

- **Thesis:** AI-Assisted Analysis and Prediction of At-Risk Diabetic Individuals (final year research project graded A). Supervisors: Dr Maleeha Azam and Dr Muhammad Umar Khan.

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

RESEARCH EXPERIENCE

University of Bradford

Jan 2025 - Sep 2025

Research Assistant, Machine Learning & LLMs

Bradford, UK

- Conducted applied ML research with responsibilities spanning independent experimental design, systematic benchmarking and analytical evaluation of model performance.
- Built graph-augmented representation learning pipelines with cross-encoder re-ranking for knowledge-grounded question answering, including large-scale dataset preprocessing and statistical modelling.
- Developed reproducible experimental pipelines and analysed model interpretability using feature attribution techniques.

COMSATS University Islamabad

Jul 2023 - Jul 2024

Research Assistant, Data Science

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk prediction on a 253,680-record CDC BRFSS dataset; Random Forest reached 93.2% accuracy, 0.98 sensitivity and 0.99 AUC after comparing random over-sampling, SMOTE and ADASYN for class imbalance.
- Analysed 20+ demographic, lifestyle and clinical indicators, identifying age, general health, BMI, blood pressure and income as the strongest diabetes correlates, and deployed the model as a public risk-screening web app (React, Flask).
- Led a first-author comparative study of sleep-efficiency prediction across four regression models (best model Random Forest, R^2 0.86), published at ICSMAI 2024 (Springer).

INDUSTRY EXPERIENCE

Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer

Leeds, UK (Remote)

- Developed backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure.

TECHNICAL SKILLS

Machine Learning & NLP : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, cross-encoder re-ranking, RAGAS evaluation, predictive modeling, ensemble methods, PyTorch, TensorFlow, scikit-learn

Knowledge & Data : Neo4j knowledge graphs, vector embeddings, semantic search, PostgreSQL, pandas, NumPy

Programming & Tools : Python, C++, SQL, FastAPI, Docker, Git, Linux, Ollama

REFEREES

Dr Irfan Mehmood, Associate Professor in Business Analytics and Artificial Intelligence, University of Bradford
i.mehmood@bradford.ac.uk | +44 (0)1274 232646

Dr Tillal Eldabi, University of Bradford
t.a.eldabi@bradford.ac.uk (*further referees available on request*)

LANGUAGES: English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)